INCH-POUND
A-A-55100A
April 5, 2023
SUPERSEDING
A-A-55100
November 10, 1992

COMMERCIAL ITEM DESCRIPTION

YARN: COTTON

The General Services Administration has authorized the use of this Commercial Item Description (CID) as a replacement for MIL-Y-16698 for all Federal agencies.

1. SCOPE. This commercial item description covers the requirements for a cotton yarn suitable for carpet yarns used in hand weaving occupational therapy.

2. CLASSIFICATION. The yarn shall be of one (1) type made of cotton fiber.

3. SALIENT CHARACTERISTICS.

3.1 Material. The yarn shall be made of clean cotton which has been carded, spun, and twisted into yarn. The yarn shall be 8/4 ply (± 3) percent, 6 to 7-1/2 turns per inch of "S" twist in the ply. (See Table I).

3.1.1 Color. The color of the yarn shall be natural, bleached white or colored as specified in the contract or procurement document. The finished yarn shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced. When white is specified, the yarn shall be bleached using a peroxide type bleach. No tinting is permissible and the use of optical bleaches is not permitted. The color cloth shall meet the colorfastness requirements specified in Table I.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Attn: DLA Troop Support, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil/.

FSC 8310

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:42Z
Check the source to verify that this is the current version before use.

3.1.1.1. Visual shade matching. The color and appearance of the yarn shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 7.4.1) with a primary light source simulating the spectral quality of average daylight, CIE Illuminant D65, with a color temperature of 6500 (± 200) Kelvin (K) illumination of 100 (± 20) foot candles. Alternatively, the color and appearance of the yarn shall match the standard sample with a primary light source simulating artificial daylight, CIE Illuminant D75, with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles in lieu of D65, (see 7.4.2). The yarn shall also be a good match to the standard sample with a secondary source simulating the spectral quality of incandescent lamplight, CIE Illuminant A, with a color temperature of 2856K (± 200).

3.2 Requirements. The finished yarn shall conform to the physical and colorfastness requirements specified in Table I when tested in accordance with test methods in Table I.

TABLE I. Physical and colorfastness requirements.

Characteristic	Requirement	Test Method
Fiber identification Cotton	See 3.1	AATCC TM20 (see 7.5)
Yarn ply	See 3.1	Visual 1/
Twist turns per inch	See 3.1	ASTM D1423 2/
Twist direction	See 3.1	ASTM D1423
Breaking strength, pounds, (min.)	7.0	ASTM D2256
Colorfastness to laundering, grade, (min.)	3-4	AATCC TM61, Test 3A 3/

1/ One determination per sample unit and results reported as “pass” or “fail”.
2/ Twist contraction is not required.
3/ Rated using the AATCC EP1 and AATCC EP2.

3.2.1 Labile sulfur. The use of dyes and compounds containing sulfur capable of oxidation to sulfuric acid is prohibited. The cloth shall be “Free” of labile sulfur when tested as specified in 3.2.1.1.

3.2.1.1 Presence of labile sulfur test. Two (2) 1.50 (± 0.01) gram samples from each sample unit shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. Prepare an acidic tin(II) chloride (stannous chloride) solution by dissolving 100 g $SnCl_2$ (ACS grade) in 100 mL of 35% hydrochloric acid (ACS grade) and adding the mixture to 50 mL of distilled water. Prepare a lead acetate solution by dissolving 5.0 g of lead acetate (CP reagent grade) in a small amount of distilled water and diluting to a total of 100 mL. If the solution is not clear, add a few drops (one at a time) of glacial acetic acid until the solution is clear. The samples shall be submersed the acidic tin(II) chloride solution. A filter paper wet out with the lead acetate solution shall be placed over the top of the test tube. The test tube containing the textile sample, tin(II) chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:42Z
Check the source to verify that this is the current version before use.

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

The rating shall be recorded. The results shall be recorded as “pass” or “fail.

3.4 Length and put-up. For Government procurement only, unless otherwise specified (see 7.6), the yarn shall be put up on spools containing 840 (± 10) yards.

3.5 Fiber identification. Each spool of finished yarn shall be labeled or ticketed for fiber content in accordance with the Title16 CFR Part 303, Rules and Regulations under the Textile Fiber Products Identification Act. (See 7.1.2)

3.6 Workmanship. The finished yarn shall conform to the quality of the product established in this document. The yarn shall be free from defects which would affect its serviceability, appearance, uniformity, and quality assurance standards defined by the technical data in the procurement documents.

4. REGULATORY REQUIREMENTS. Unless otherwise specified, the offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403, 23.404(b) of the Federal Acquisition Regulation FAR, or the sustainable acquisition of EO 13693; as applicable, (see 7.1.3).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The yarn shall meet the salient characteristics of this CID, conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

5.2 Market acceptability. The finished yarn offered must have been sold to the government or commercial market.

5.3 Inspection procedures. Sampling for inspection shall be conducted in accordance with ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection of Attributes (see 7.2.2).

5.4 Visual examination. The finished yarn shall be examined for the defects listed below that are clearly visible at normal inspection distance of approximately three (3) feet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:42Z
Check the source to verify that this is the current version before use.

TABLE II. Visual examination defects.

Material defects: Any spot, stain, burr, or shive. Color not as specified, off shade, not uniform. Foreign matter or objectionable odor.
Label: Label missing, incorrect, or illegible. Required information missing or not in accordance with the contract or procurement documents.
Packaging: Not packaged in accordance with the contract or procurement documents.

5.5 Acceptance criteria. Acceptance criteria shall be as specified in the contract or procurement document (see 7.6).

6. PACKAGING. The preservation, packing and marking shall be as specified in the contract or order (see 7.6).

7. NOTES.

7.1 Source of Government documents:

7.1.1 Copies of Government documents are available online at https://quicksearch.dla.mil.

7.1.2 Title 16 CFR Part 303, Rules and Regulations under the Textile Fiber Products Identification Act are available online at https://www.ecfr.gov.

7.1.3 EPA and EO 13693 information is available online at https://www.epa.gov.

7.1.4 Federal Acquisition Regulations (FAR) are available online at https://acquisition.gov.

7.2 Source of non-Government documents.

7.2.1 AATCC test methods are available for purchase online at https://aatcc.org.

7.2.2 ASQ/ANSI Z1.4 – Sampling Procedures are available online at https://www.asq.org.

7.2.3 ASTM Standards are available for purchase online at https://www.astm.org.

7.3 Intended use. The yarn is used in hand weaving for occupational therapy.

7.4 Standard samples. For access to standard shade samples of yarn, address the contracting activity issuing the invitation for bids or request for proposal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:42Z
Check the source to verify that this is the current version before use.

7.4.1 Visual shade matching. In 2017, Option A of AATCC EP9 was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5º) angle relative to the horizontal. The observer will view the specimens at a 90º angle, relative to the plane of the specimens".

7.4.2 In 2022, the U.S. Military (All Services) accepted D65 as the new preferred light source for visual shade matching due to the supply chain issues with D75.

7.5 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for a stated requirement.

7.6 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this Commercial Item Description.
b. Color required (see 3.1.1).
c. Length and put-up (see.3.4).
d. Product conformance provisions (see 5.1).
e. Acceptance criteria (see 5.5).
f. Packaging requirements (see 6.).

7.7 Key words.

Occupational therapy
Twist

MILITARY INTERESTS:

CIVIL AGENCY COORDINATING ACTVITY:
GSA-FAS

Custodians:
Army - GL

Preparing Activity
DLA-CT

Review activities:
Army - MD

(Project: 8310-2023-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:42Z
Check the source to verify that this is the current version before use.